Ferromagnetic Semiconductors

Technology, Physics

Spintronics

Conventional electronics:

charge of electron

used to achieve functionalities – diodes, transistors, detectors, lasers…

Spintronics:

manipulate electron spin (or resulting magnetism)

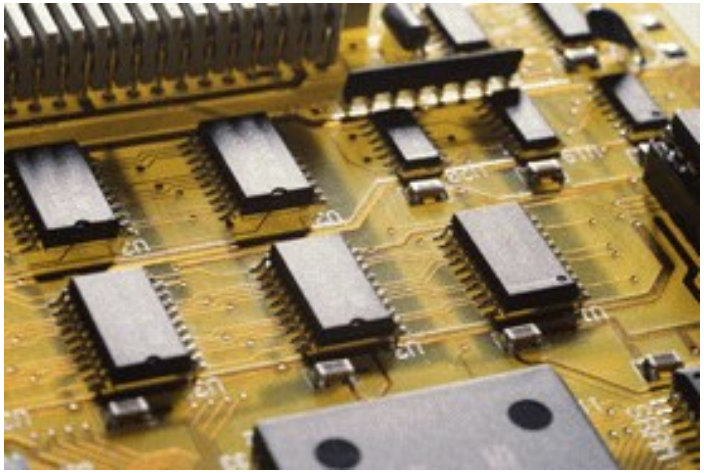
to achieve new/improved functionalities - spin transistors, memories, higher speed, lower power, tunable detectors and lasers, bits (Q-bits) for quantum computing….

Why magnetic semiconductors?

Possible applications

Nearly incompatible technologies in present-day computers:

semiconductors: processing



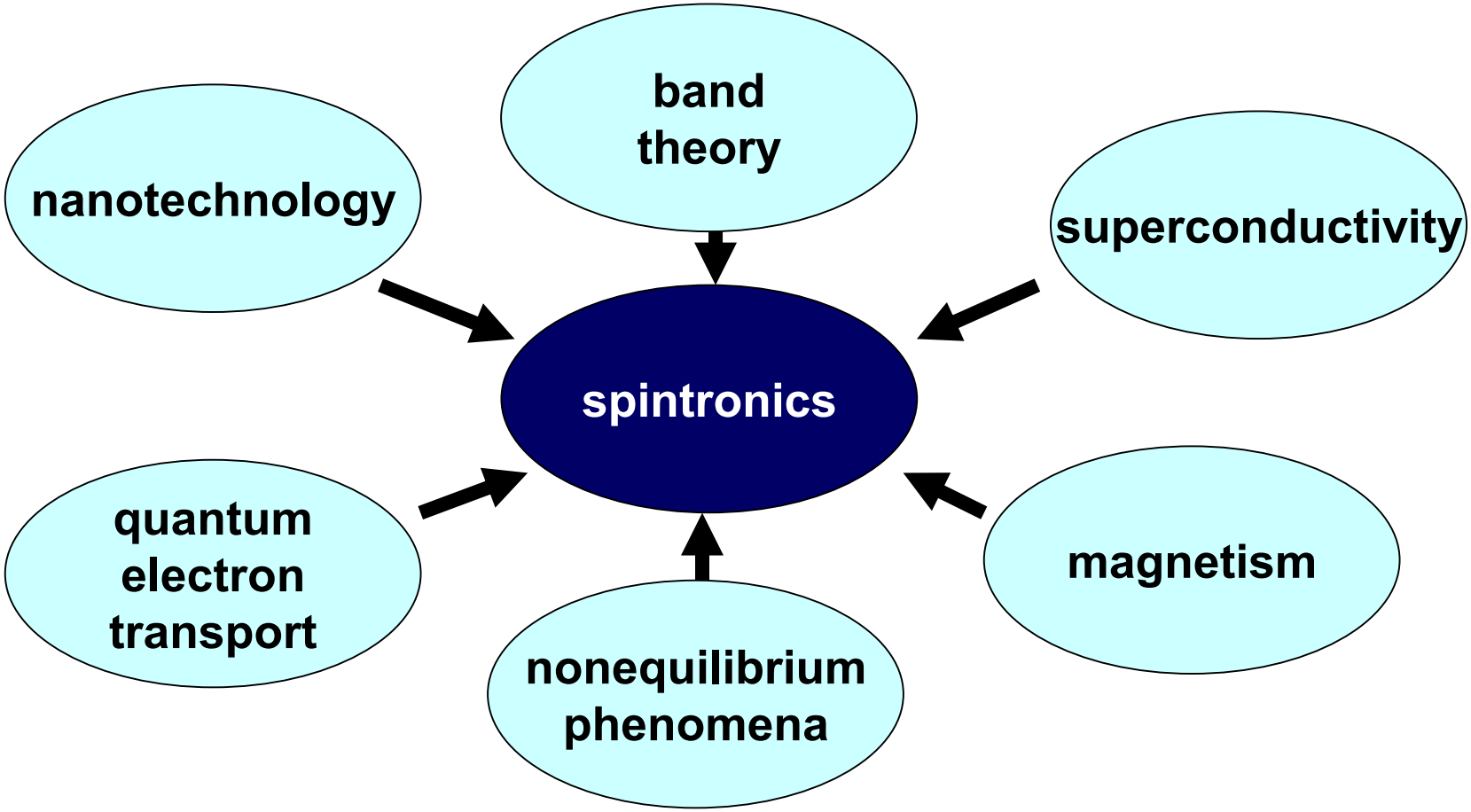
ferromagnets: data storage



ferromagnetic semiconductors: integration on a single chip?

single-chip computers for embedded applications:
cell phones, intelligent appliances, security

Multidisciplinary Field



Material Structures

Growing Stage

FM / Conductors / FM

GMR : read head, sensor, MRAM

Developing Stage

FM / Insulators / FM

TMR : MRAM, read head, logic device

Fundamental Stage

FM / Semiconductors / FM

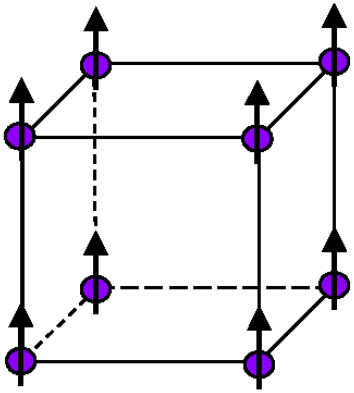
spin-FET, spin-LED, logic device

Fundamental Stage

Semiconductor / DMS / Semiconductor

spin- RTD, logic device, quantum computing

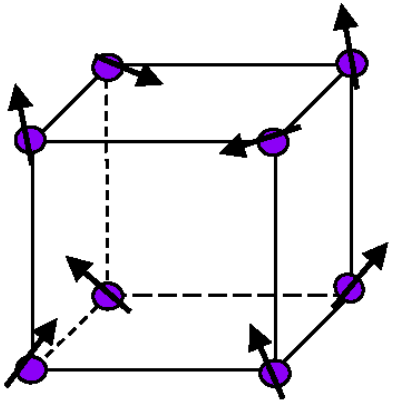
Ferromagnetism - Paramagnetism



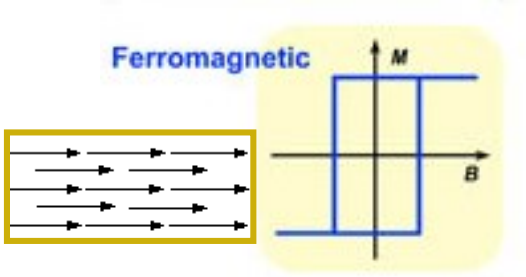
Ferromagnetism
(magnetic)

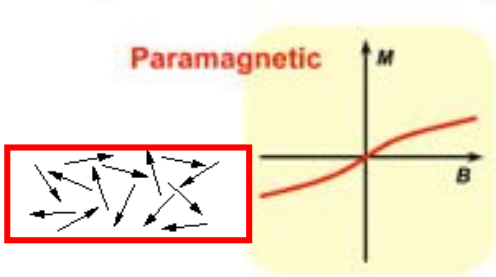
above
T_{Currie}

$T > T_c$



Paramagnetism
(nonmagnetic)



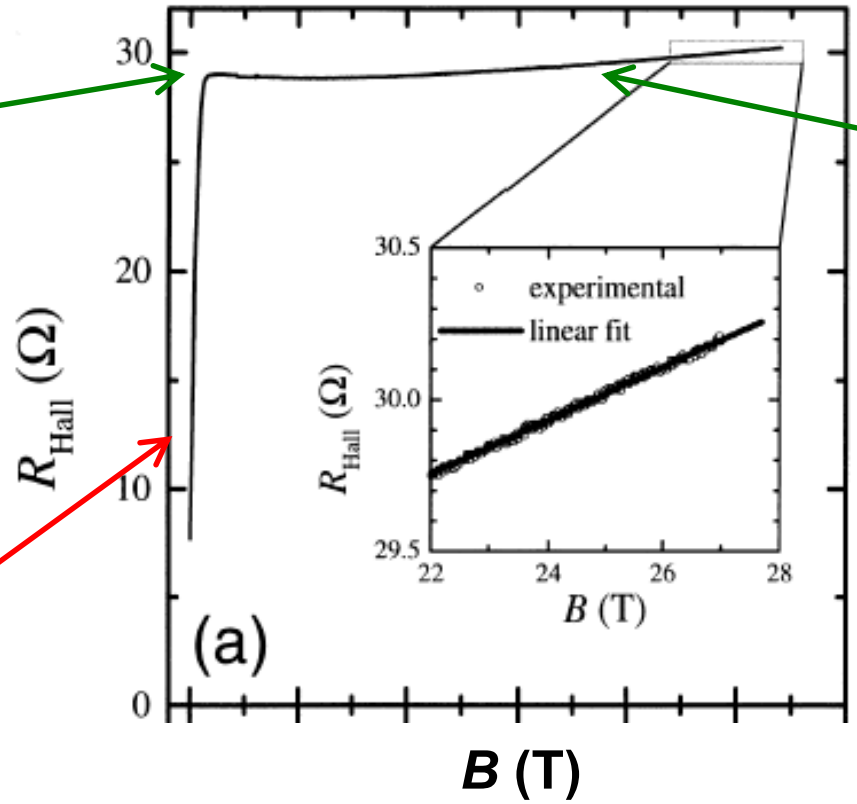


Anomalous Hall effect

Hall effect in the *absence* of an applied magnetic field
(in itinerant ferromagnets, due to spin-orbit coupling)

saturation of magnetization

anomalous Hall effect



normal Hall effect: roughly linear in B

(R_H / B)

Omiya *et al.*, Physica E **7**, 976 (2000)

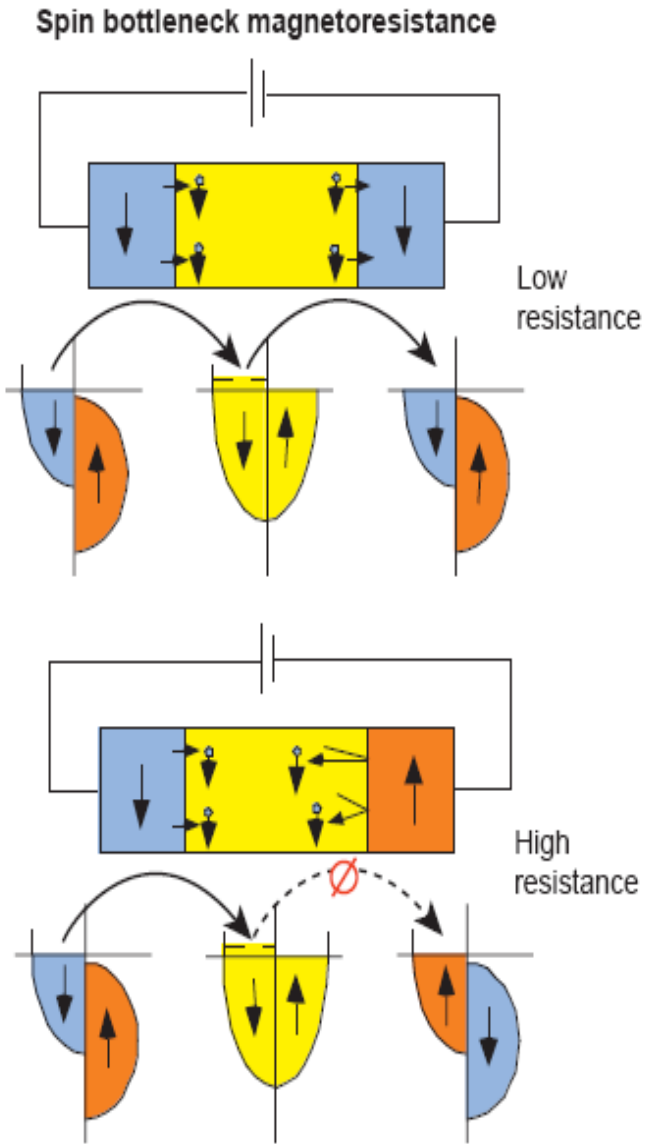
1988: discovery of GMR

Giant Magnetic Resistance:

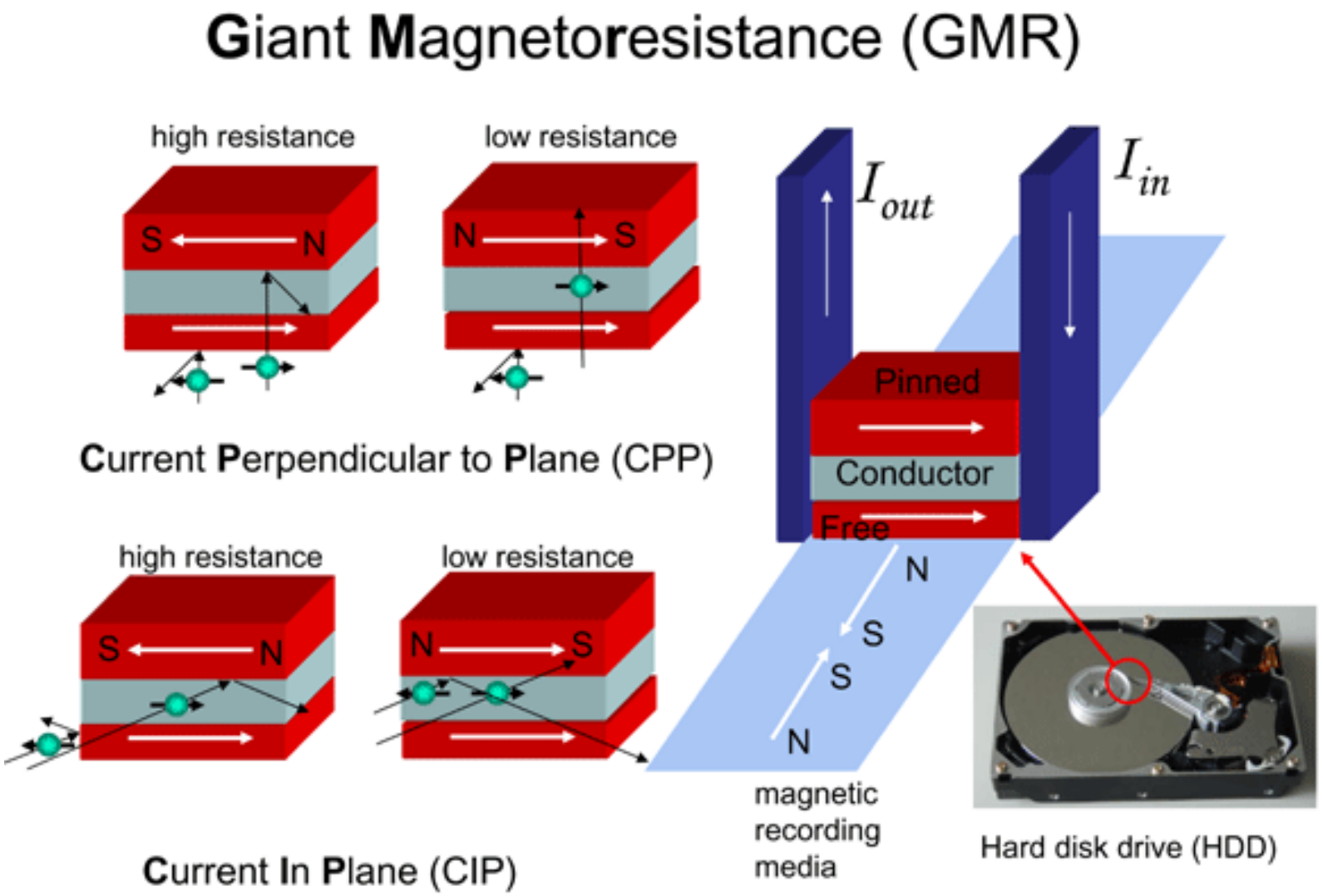
In alternate FM/nonmagnetic layered system,

R is low when the magnetic moments in the FM layers are aligned,

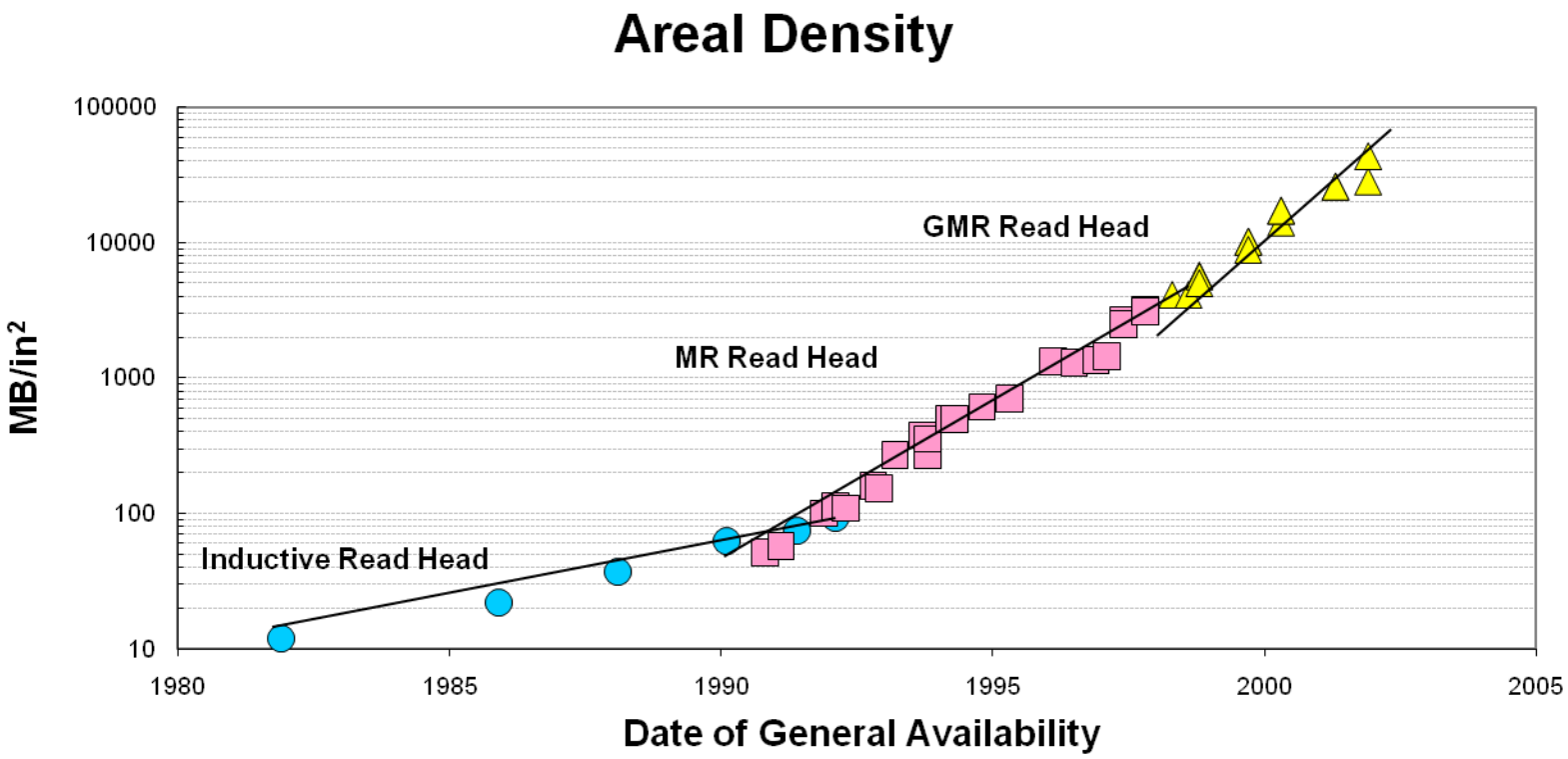
R is high when the magnetic moments in the FM layers are antialigned.



How the Read Head Retrieve Data



Hard Drive Information Density



Magnetoresistive Random Access Memory (MRAM).

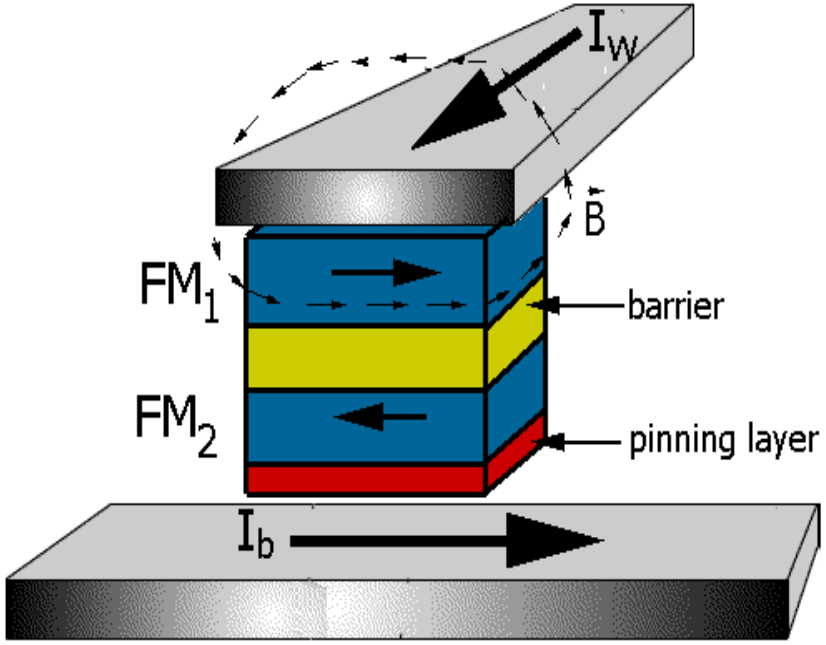
- huge enhancements of speed, reliability, fabrication, and power consumption
- two ferromagnetic (FM) films separated by a non-magnetic spacer.
- When the FM layers are parallel,

an channel exists for minority spin electrons to conduct from one layer to the other ferromagnetic layer (low resistance state.)

- when the FM layers are anti-aligned

the resistance is high.

- To store data bits the magnetic field from the wire reverses one of the FM layers.
- one of the FM layers is pinned by placing an biased layer directly underneath one of the layers, to make it more difficult to switch the magnetic moment.



MRAM

fast, small, low-power, durable, and non-volatile

Table I. Comparison of MRAM Expected Features With Other Memory Technologies[a]

	SRAM	DRAM	FLASH	FRAM
Read	Fast	Moderate	Fast	Moderate
Write	Fast	Moderate	**Slow**	Moderate
Nonvolatile	**No**	**No**	Yes	**Partially**[b]
Endurance	Unlimited	Unlimited	**Limited**[c]	**Limited**[b]
Refresh	No	**Yes**	No	No
Cell size	**Large**	Small	Small	Medium
Low voltage	Yes	Limited	**No**	Limited

[a] Bold letters indicate undesirable attributes.
[b] Destructive read and limited read/write endurance.
[c] Limited write endurance.

2006- First commercial 4Mb MRAM

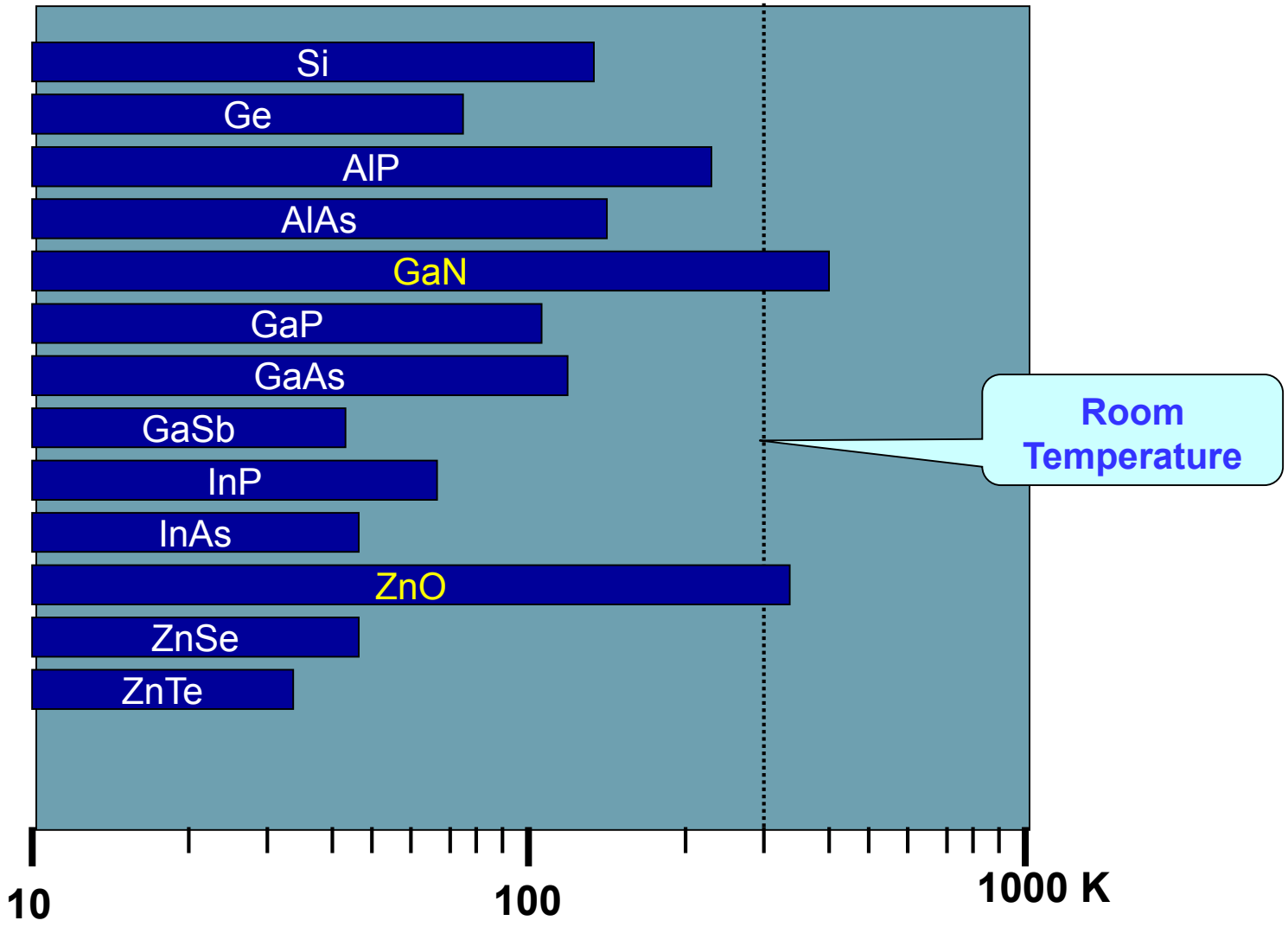


RAM chip that actually won't forget → instant on-and-off computers

“History” of magnetic semiconductors

Material	Year	T_c	Transport
EuO, EuS	1960-70	<65 K	Insulating
(II,Mn)VI	1970-80	0	Semiconductor
PbSnMnTe CdTeMn:N ZnMnTe:X	1980-1990	<1.5 K	P-type semiconductor
InMnAs	1992	7 K	Insulator
(Ga,Mn)As	1996	110 K	Semiconductor, bad metal
GaMnN MnGe MnZnO CrZnTe	2001 NCSU 2003 Univ of Minn 2003 Sweden 2003 Japan	400K 350K 450K 300K	Semiconductor

Predicted Curie Temperatures



Dietl et al., Science , (2000)

Diluted Magnetic Semiconductors

- magnetic materials closely lattice-matched to GaAs are of special interest due to their potential of a seamless integration with well established standard materials
- III-V based material composition, doped with Mn, is generally accepted as a promising system.
- The (III,Mn)As semiconductors (with Ga, In, Al as cations) reached Curie temperatures in excess of 100 K and have prospect of a further increase.

III-V DMS

GaAs, InAs and their alloy + Mn.
Mn substitute the cation and introduce a hole.

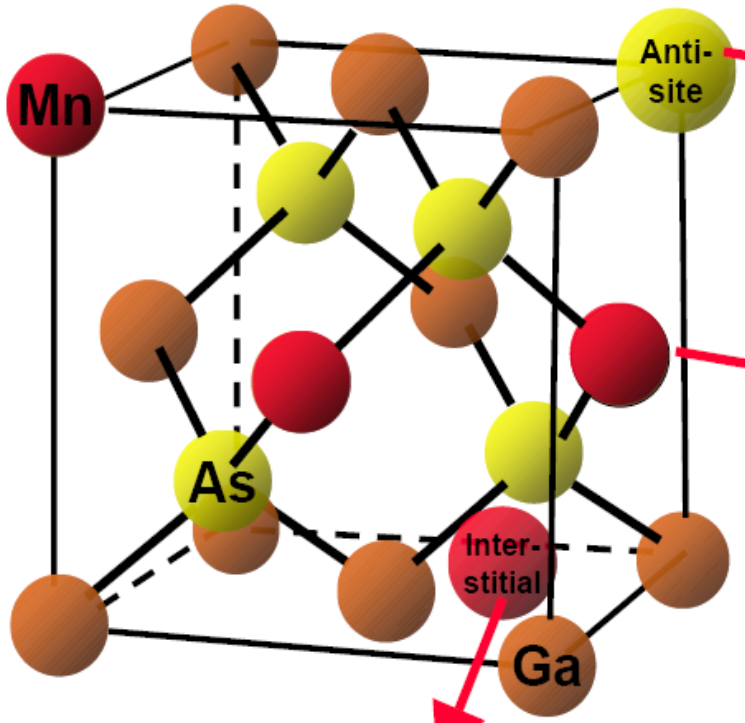
Low solubility of the magnetic element,
under normal growth condition.

→ Non-equilibrium epitaxial growth methods (MBE)
to overcome the thermodynamic solubility limit.
Standard MBE growth condition not sufficiently far from equilibrium

→ Low temperature MBE

1992 FM InMnAs
1996 FM GaMnAs

GaMnAs structure



Anti-site As defects in low temperature GaAs
Double donor
Q=+2e

Substitutional Mn:
Q=-e and local 5/2 moment

Interstitial Mn: Double donor Q=+2e

To increase T_C one has to

- Minimize As antisite defects
- Minimize interstitial Mn
- Get sufficiently high Mn content

As antisite

As flux reduced to the minimum necessary in order to maintain a 2D RHEED pattern at the selected temperature.

2 Ga cell to maintain the exact stoichiometry during both GaAs and GaMnAs growth.

Use of As_2 instead of As_4

Interstitial Mn

Interstitial Mn are detrimental for FM:

- are double donor
- are attracted by substitutional Mn and coupled with them antiferromagnetically → reduce the effective Mn moments concentration x_{eff}

Evidences (by RBS and PIXE) of the presence of interstitial Mn in as grown GaMnAs.

<u>Low T annealing</u> reduce the interstitials density that diffuse toward the surface, rise T_C and p

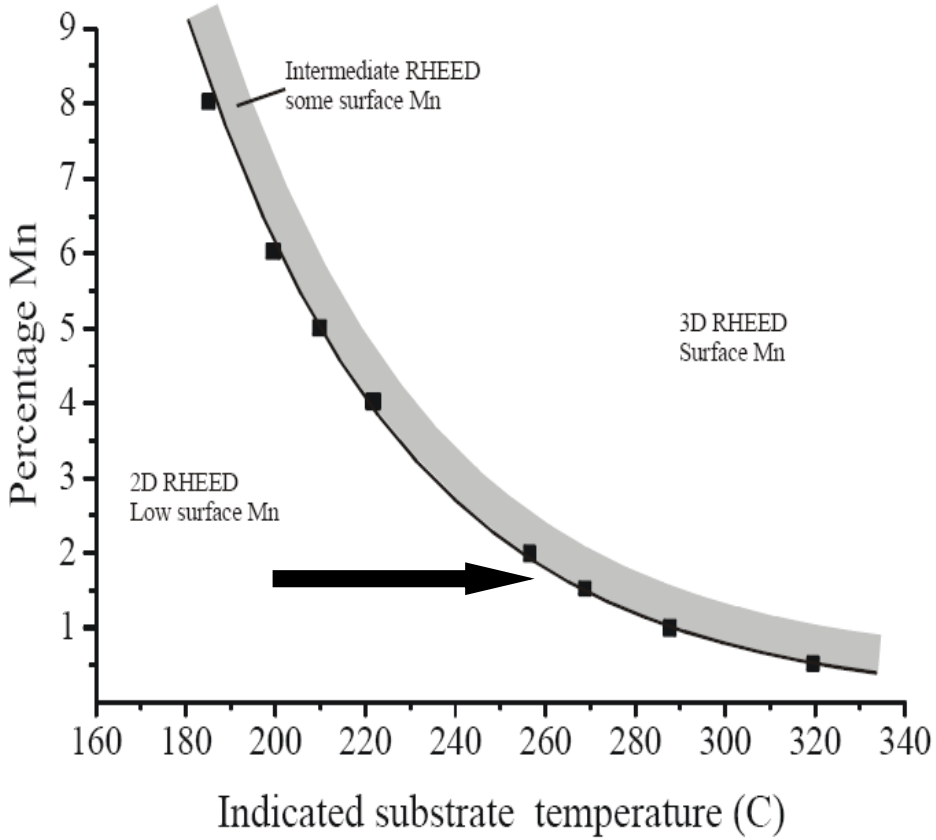
Yu *et al*, PRB 65,201303R (02)

Edmonds *et al*, PRL 92, 037201 (04)

Mn incorporation

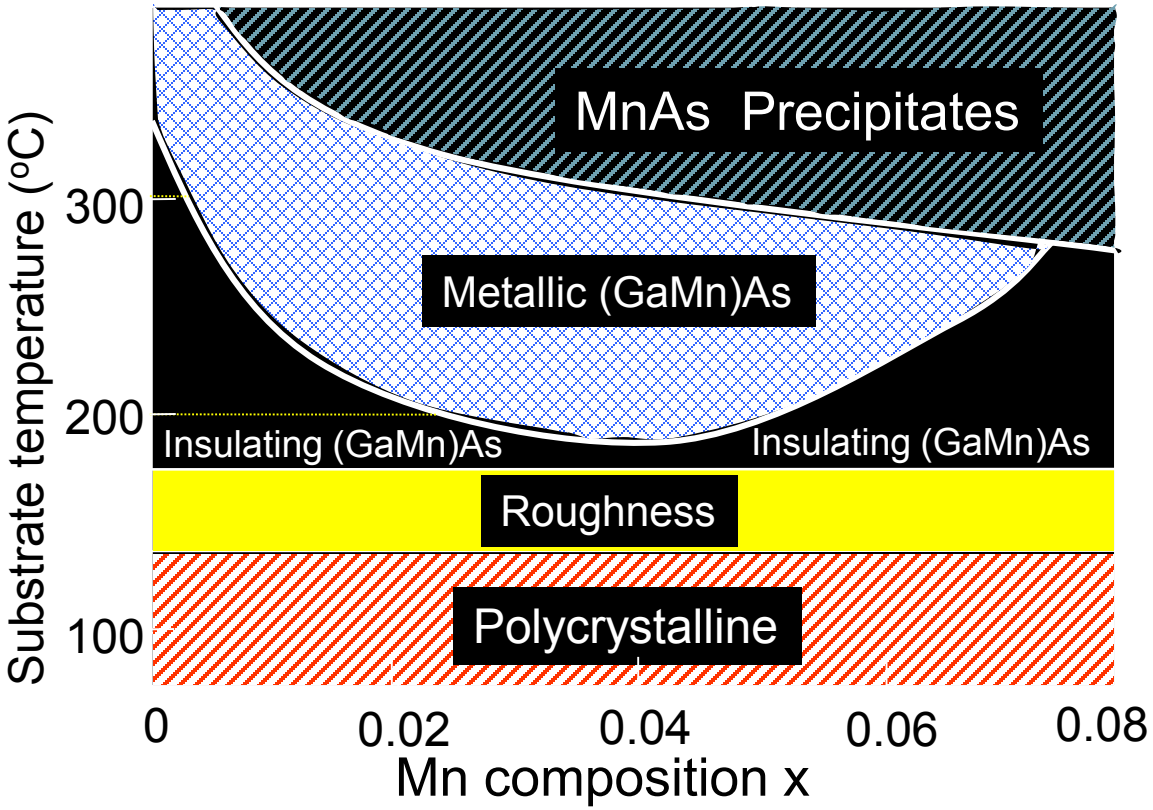
To increase Mn content and minimize surface segregation,
low growth temperature

Ideal temperature vs Mn content identified by monitoring the RHEED :
the highest T giving 2D RHEED



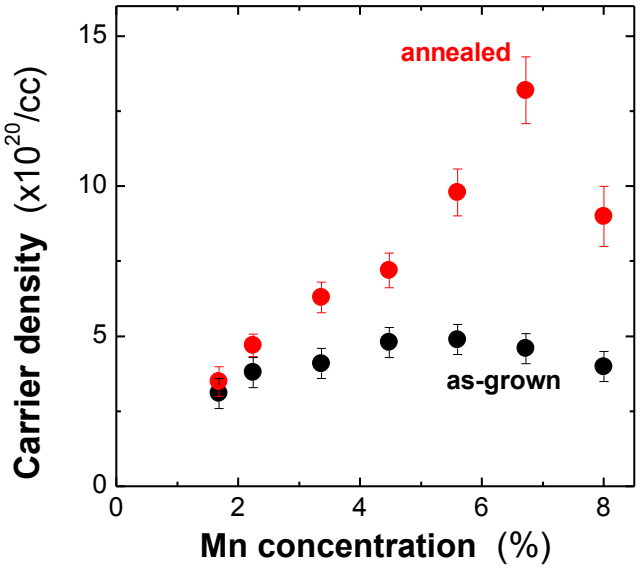
R.P. Campion et al, JCG 251, 311 (03)

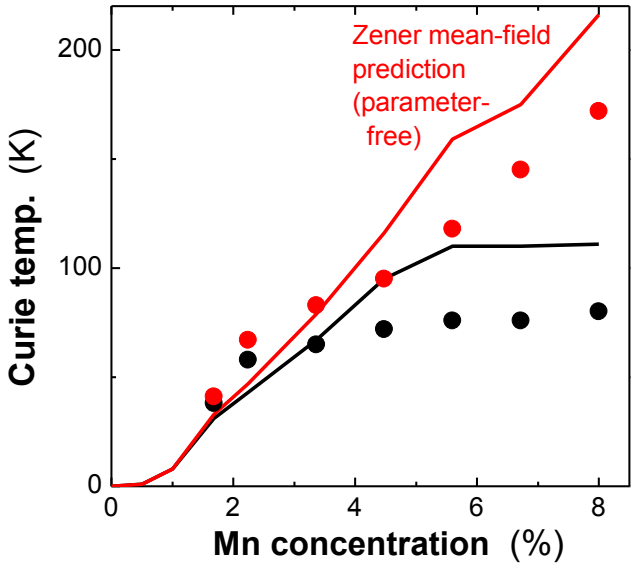
MBE Phase Diagram – $Ga_{1-x}Mn_xAs$



H. Ohno , J. of Magnetism and Magnetic Materials **200** (1999)

Curie temperatures





Max. T_C~180K

Wang et al., JAP '04

FM in Mn based Zincblend DMS

• Antiferromagnetic direct coupling between Mn ions. Dominate in undoped materials.

• Ferromagnetic coupling in p-type materials as a result of exchange interaction between substitutional Mn S=5/2 and hole spins.
The exchange interaction follows from hybridization between Mn *d* orbital and valence band *p* orbital.

→ Hole mediated FM

PRB 72, 165204(05)

Hole mediated FM

In a mean field virtual crystal approximation

$$T_C \propto x p^{1/3}$$

x = substitutional Mn
p = hole density

In III-V DMS the holes comes from Mn !!!

→ x and p are intimately related

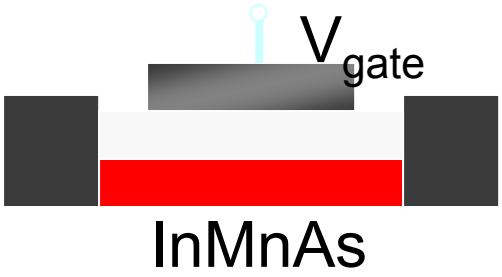
Room temperature T_C is expected for $Ga_{0.9}Mn_{0.1}As$.

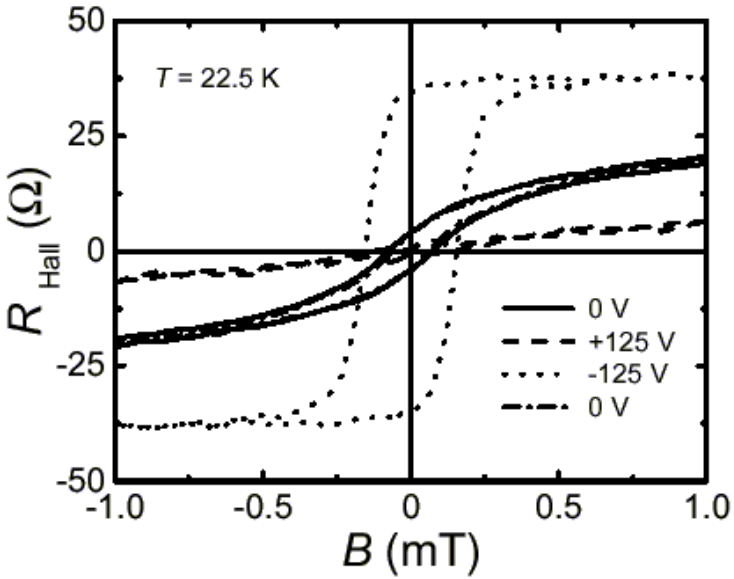
PRB 72, 165204(05)

Carrier-mediated ferromagnetism

Spin-FET

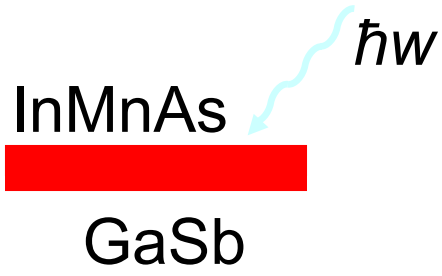
H. Ohno et al., Nature (2000)

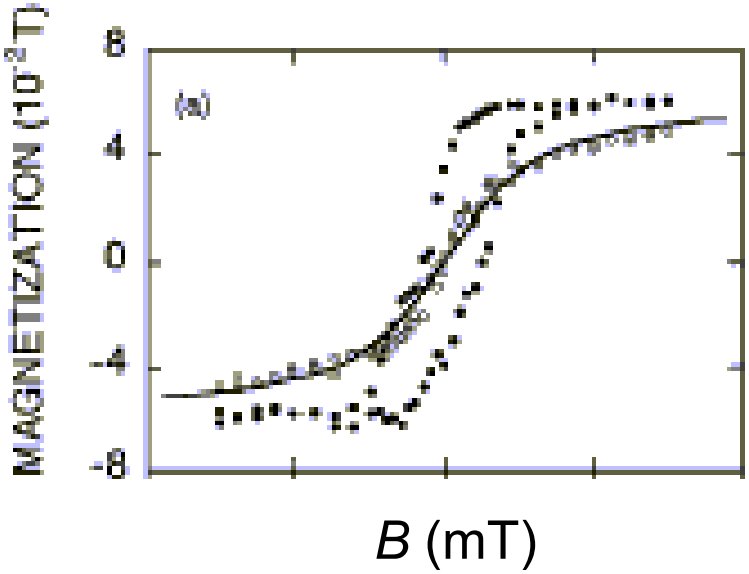




Photogenerated magnetism

Koshihara PRL (1997)





$III_{1-x}\ Mn_x\ V$ for Spintronics

Random Alloys

- Precipitates for high Mn concentration
- Poor Structural/Optical Properties

New Approach

- δ-doping highly effective for increasing concentration
- "Digital alloys" result in high quality materials

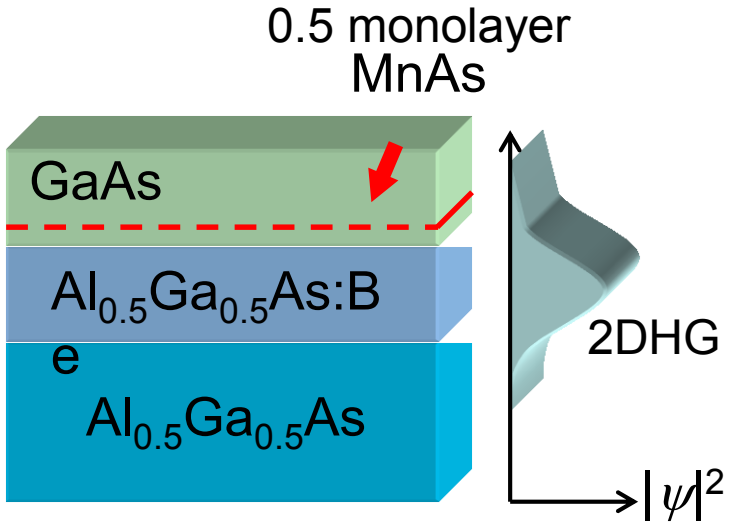
III-V/Mn and III-V/GaMn "digital alloys"

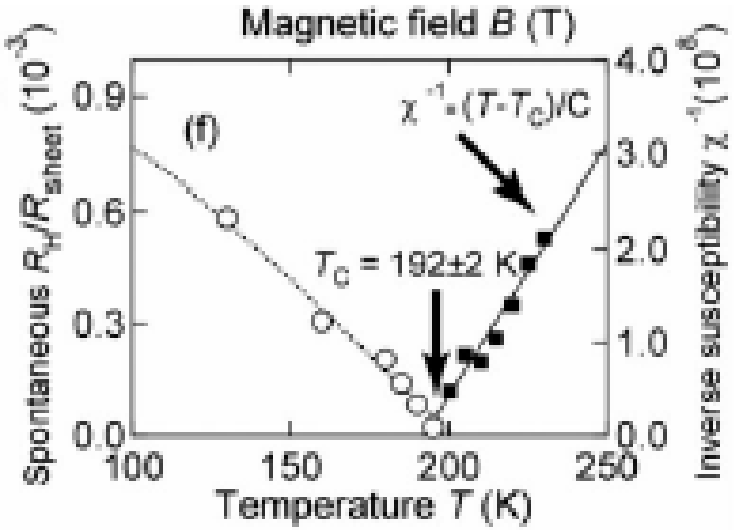
Outcomes :

- Increased Mn concentration and improved structural quality
- Quasi 2-D carrier system in region of ion spins – enhanced T_C
- Improved Optical and transport properties (ordered alloy)

p-doped (Ga,Mn)As δ-doped layer

Nazmul *et al.*, PRL **95**, 017201 (2005)





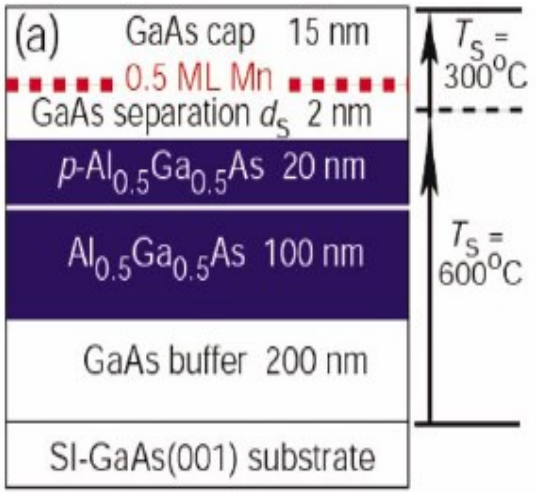
allows higher local concentration of Mn

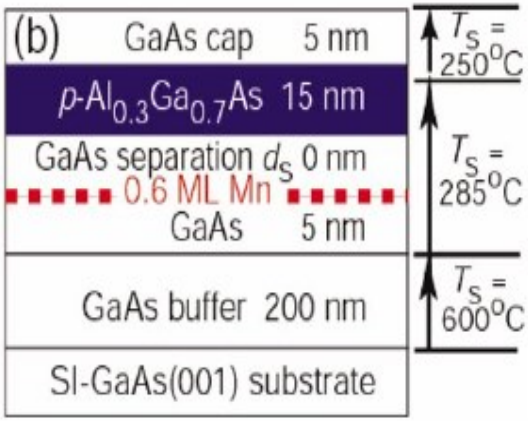
tail of hole concentration of 2DHG in δ layer

T_c up to 250 K

quasi-two-dimensional ferromagnet (interdiffusion?)

Mn δ-doping and heterostructue design





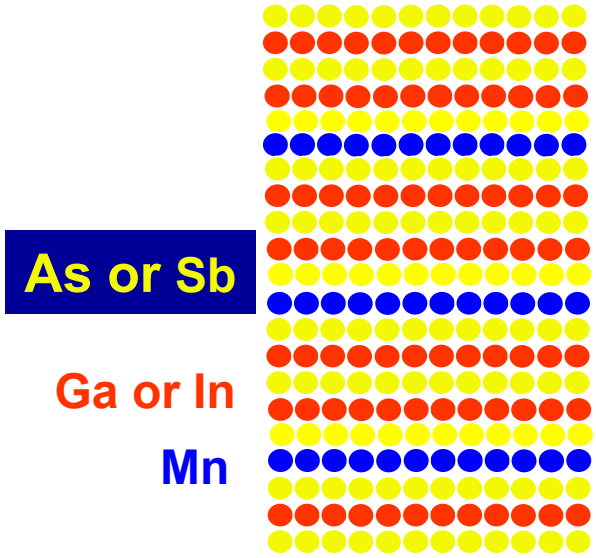
Record T_C = 190 K after annealing

Record T_C = 250 K after annealing !

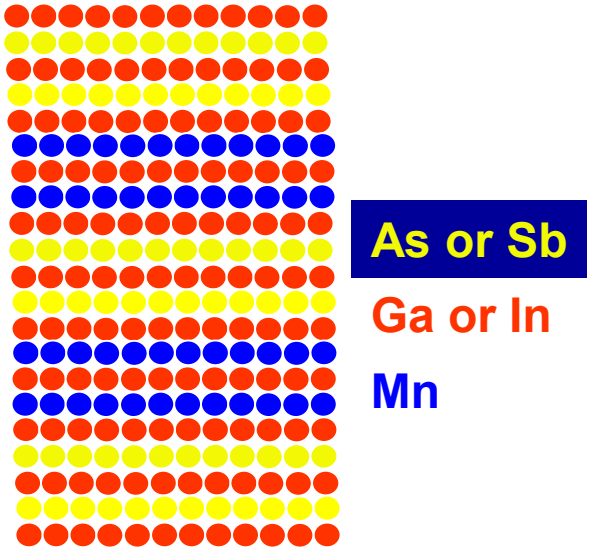
E_F effect on Mn interstitial density?

Nazmul *et al*, PRL 95, 017201 (05)

Digital alloys – 2 Types

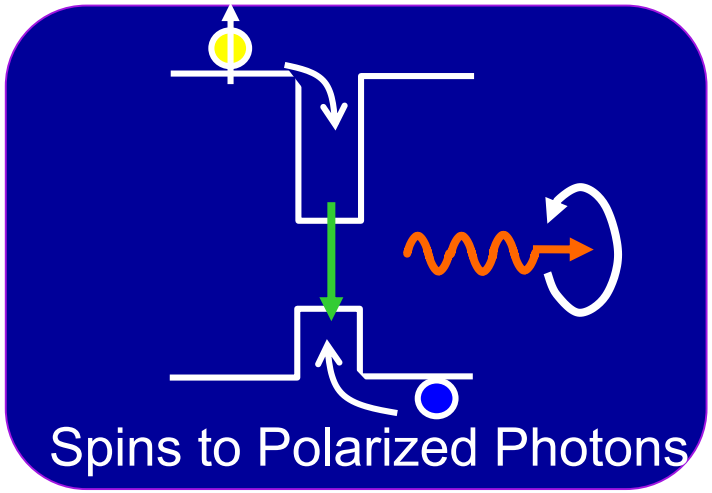


GaAs:Mn



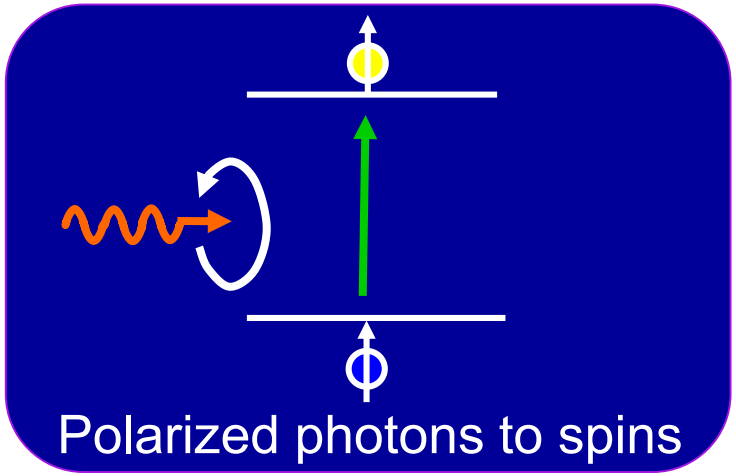
GaAs:GaMn

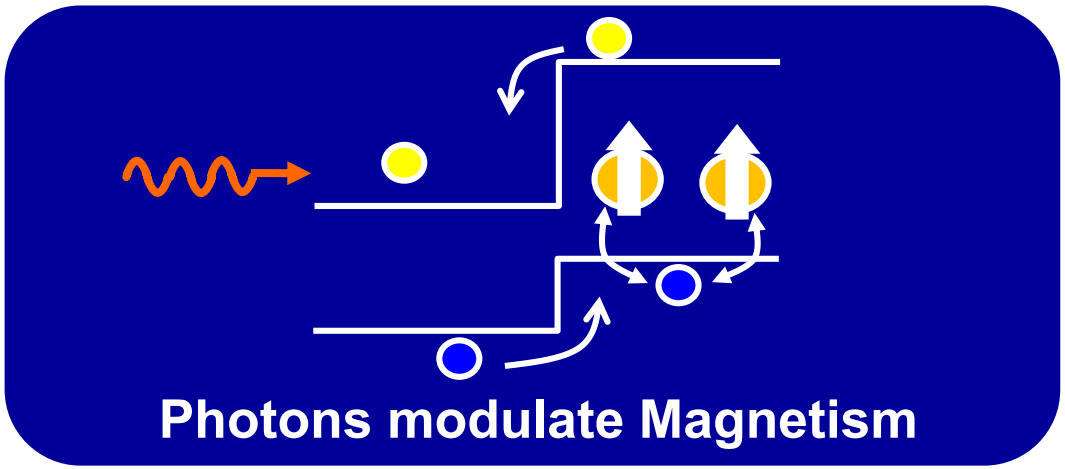
Building Blocks for Spin-Photonics



spin-LED

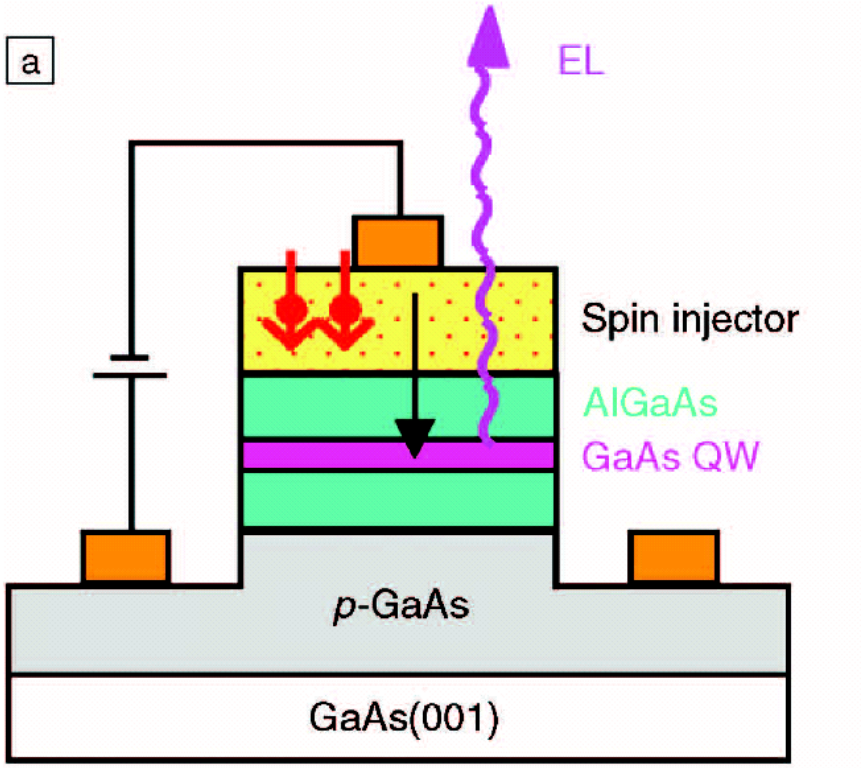
optically controlled
ferromagnetism





Spin-LED

Spin-polarised carriers injected from the contact emit circularly polarised light



Light Circular Polarization

ZnMnSe LED
circular polarization 72%
for B = 4T

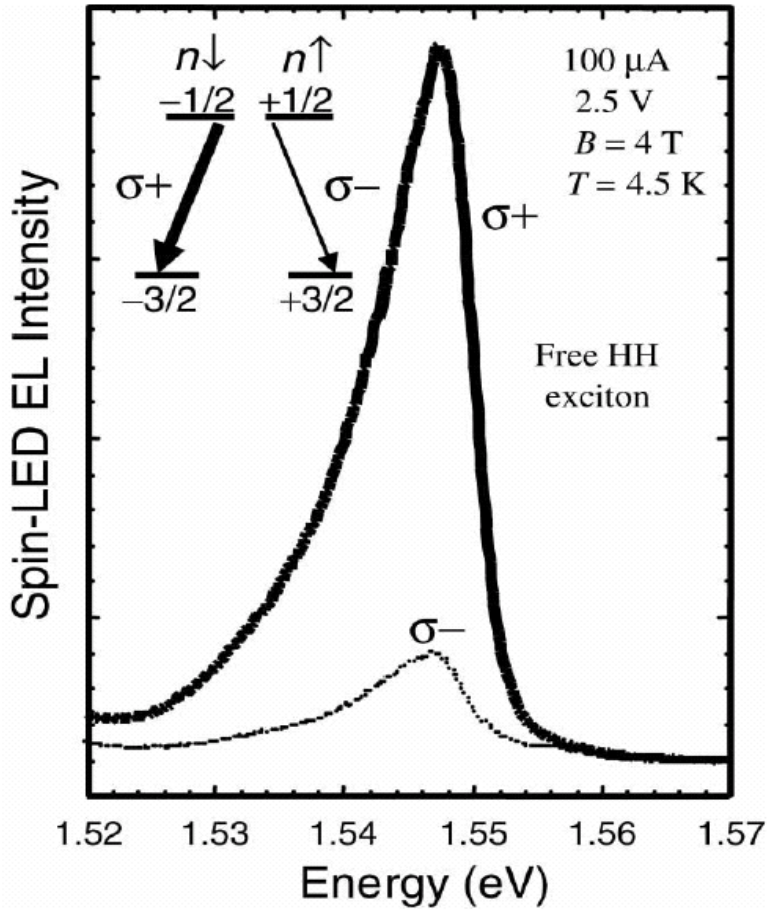
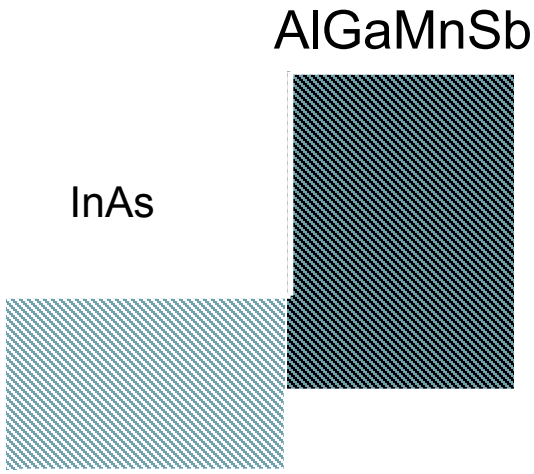
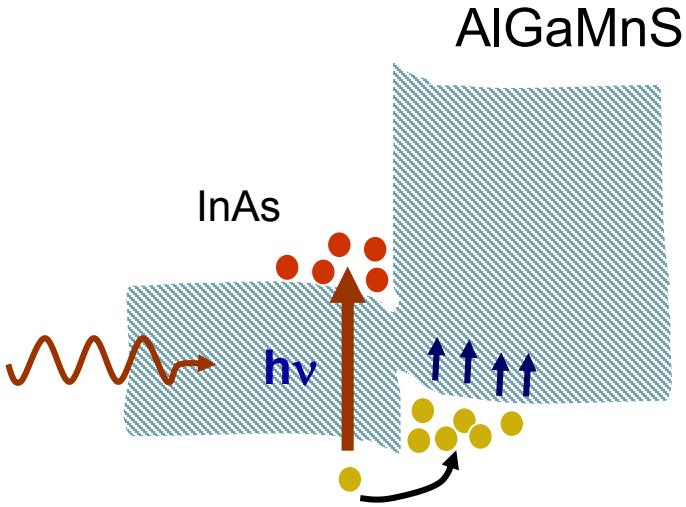


Photo-induced Ferromagnetism

- the presence of light drive the paramagnetic ions into ferromagnetic order
- Heterostructure Band offsets ➡ Holes go to Incipient Ferromagnetic Material
  Mechanism for Ferromagnetism is Carrier-induced



No Illumination



E_g (InAs) < hν < E_g (AlGaMnSb)

Type-II Band Alignment

Munekata *et al*, PRL **78,** 4617 (1997) (InMnAs)

Semiconductor spintronics

- an emerging field of applied science
- aims at integrating the functionalities of semiconductor and magnetic materials
- developing new types of devices for information storage and processing.
- magnetic properties of ferromagnetic semiconductors can be exploited for carrier spin manipulation
- retaining a compatibility with existing electronics

Advantages

- nonvolatility,
- increased data processing speed,
- decreased electric power consumption,
- increased integration densities compared with conventional semiconductor devices.

Major Challenges

- achieving a room temperature ferromagnetism in a proper semiconductor material,
- optimisation of electron spin lifetimes there,
- detection of spin coherence in nanoscale structures,
- transport of spin-polarised carriers across relevant length scales and heterointerfaces,
- the manipulation of both electron and nuclear spins on sufficiently fast time scales.